INCH-POUND
MIL-W-16878/34A(NAVY)
11 September 1992
SUPERSEDING
MIL-W-16878/34(NAVY)
25 June 1982

MILITARY SPECIFICATION SHEET

# WIRE, ELECTRICAL, POLYTETRAFLUOROETHYLENE (PTFE) INSULATED, 200°C, 1000 VOLTS, DOUBLE WRAPPED INSULATION

This specification is approved for use by the Department of the Navy and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-W-16878.

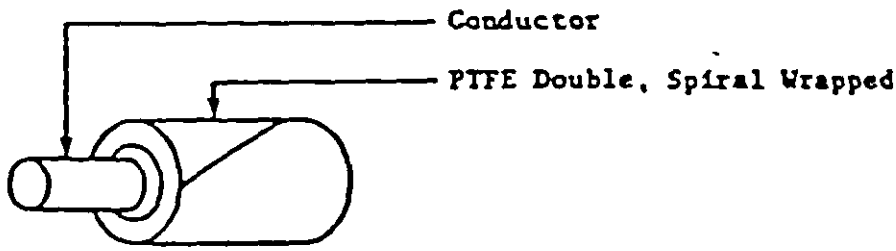



FIGURE 1. Construction.

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TABLE I. Construction details.

| PIN 1/ | Wire size | Strand-ing | Conductor | | Conductor diameter (nominal) (inch) | Finished wire diameter (inch) | |
|---|---|---|---|---|---|---|---|
| | | | Material | Coating | | Min | Max |
| M16878/34-BAB* | 32 | 7 X 40 | Copper | Silver | 0.010 | 0.036 | 0.044 |
| M16878/34-BBB* | 30 | 7 X 38 | Copper | Silver | .012 | .038 | .046 |
| M16878/34-BCB* | 28 | 7 X 36 | Copper | Silver | .015 | .041 | .049 |
| M16878/34-BDB* | 26 | 7 X 34 | Copper | Silver | .019 | .045 | .053 |
| M16878/34-BEB* | 24 | 7 X 32 | Copper | Silver | .024 | .050 | .058 |
| M16878/34-BFB* | 22 | 7 X 30 | Copper | Silver | .030 | .056 | .064 |
| M16878/34-BGB* | 20 | 7 X 28 | Copper | Silver | .038 | .064 | .072 |
| M16878/34-BHB* | 18 | 7 X 26 | Copper | Silver | .048 | .074 | .084 |
| M16878/34-BJE* | 16 | 19 X 29 | Copper | Silver | .057 | .083 | .095 |
| M16878/34-BKE* | 14 | 19 X 27 | Copper | Silver | .072 | .098 | .114 |
| M16878/34-BLE* | 12 | 19 X 25 | Copper | Silver | .091 | .117 | .133 |
| M16878/34-BMG* | 10 | 37 X 26 | Copper | Silver | .111 | .137 | .153 |
| M16878/34-BNL* | 8 | 133 X 29 | Copper | Silver | .169 | .199 | .219 |
| M16878/34-BPL* | 6 | 133 X 27 | Copper | Silver | .213 | .253 | .273 |
| M16878/34-BRL* | 4 | 133 X 25 | Copper | Silver | .257 | .287 | .307 |

1/ PIN stands for part or identifying number.

ADDITIONAL REQUIREMENTS:

Construction: The wire shall be insulated with two spirally-wrapped layers of 0.005-inch (nominal) thick PTFE tape. The first layer of tape shall be applied in the opposite direction to the lay of the conductor, and the second layer of tape shall be applied in the opposite direction of the first layer. Then the two layers shall be heat-bonded and fused.

Visual and mechanical examination: Required.

Spark test: 5.0 kV.

Impulse dielectric test: 8.0 kV.

Dielectric withstanding voltage: 3.0 kV.

Insulation resistance: $IR = K \log_{10} D/d$.

Where: IR = Minimum insulation resistance in megohms per 1000 feet at 20°C.
K = 50,000.
D = Maximum average diameter of finished wire.
d = Conductor diameter.

Conductor resistance: See table I.

Cold bend: Condition 4 hours at minus 65 ± 1°C (see table II).



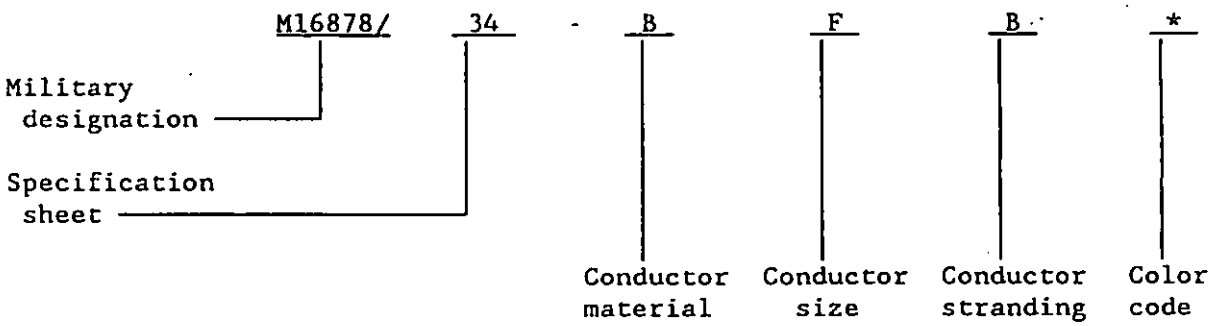
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TABLE II. Cold bend mandrel sizes.

| Wire size | Cold bend mandrel (maximum diameter in inches) |
|---|---|
| 32 through 16 | 1 |
| 14, 12 | 2 |
| 10, 8 | 3 |

Concentricity: 70 percent (minimum).
Surface resistance: Not required.
Wrap back: Required.
Heat resistance: Condition at 290°C.
Flammability: Meets requirements of UL VW1 Flame Test.
Toxicity: Meets requirements of concentration levels specified in MIL-E-917.
Shrinkage: Required.
Heat aging: Not required.
Insulation tensile strength: 4000 pounds force per square inch (minimum).
Insulation elongation: 150 percent (minimum).
Marking and stripe durability: Required.
Fungus resistance: Not required.
Maximum dc resistance of finished wire: See table I of MIL-W-16878.
Wire length requirements: See appendix B.
Identification of product: Required for wire sizes 22 AWG and larger.

PIN example (see MIL-W-16878):




Revision letters are not used to denote changes due to the extensiveness of the changes.

Preparing activity:
Navy - SH
(Project 6145-N330-33)



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